

LIBEREC, Czech Republic

WMO: 116030

Lat: 50.7699N

Lon: 15.0240E

Elev: 401

StdP: 96.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-10.4	-16.9	0.9	-12.0	-13.8	1.2	-9.0	12.2	3.8	10.4	2.1	2.0	300	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	29.4	19.1	27.4	18.2	25.5	17.5	20.1	26.7	19.2	25.5	18.3	24.0	3.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.5	22.4	17.0	12.7	21.3	16.1	12.0	20.6	59.4	26.7	56.2	25.7	53.2	24.0	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.8	6.9	DB	-16.9	32.4	4.0	2.1	-19.8	34.0	-22.1	35.2	-24.3	36.4	-27.2	37.9	(4)
(5)			WB	-17.2	22.1	4.0	0.9	-20.0	22.7	-22.3	23.2	-24.6	23.7	-27.5	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-1.5	-0.1	3.2	8.3	12.6	15.7	17.8	17.5	13.2	8.8	4.0	-0.1	(6)
(7)		DBStd	7.93	4.77	4.73	4.15	4.56	3.96	3.79	3.51	3.40	3.59	4.01	3.99	4.43	(7)
(8)		HDD10.0	1532	355	284	212	84	19	2	0	0	9	72	182	312	(8)
(9)		HDD18.3	3748	613	517	469	302	184	97	54	57	158	296	430	571	(9)
(10)		CDD10.0	922	0	0	2	33	98	173	241	233	106	34	2	0	(10)
(11)		CDD18.3	96	0	0	0	1	4	18	37	32	4	0	0	0	(11)
(12)		CDH23.3	971	0	0	0	4	46	193	394	299	34	0	0	0	(12)
(13)		CDH26.7	248	0	0	0	0	4	44	117	79	3	0	0	0	(13)
(14)	Wind	WSAvg	3.1	3.8	3.6	3.4	3.0	2.8	2.6	2.5	2.4	2.6	3.1	3.6	3.8	(14)
(15)	Precipitation	PrecAvg	946	74	63	71	49	78	95	121	108	77	62	72	74	(15)
(16)		PrecMax	1270	156	117	176	117	150	224	287	334	177	131	157	130	(16)
(17)		PrecMin	618	5	6	33	2	29	36	28	21	29	14	0	29	(17)
(18)		PrecStd	134	38	31	34	23	35	39	63	69	40	32	37	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	11.2	17.0	23.7	27.1	30.4	32.6	32.1	26.5	21.0	14.3	9.8	(19)
(20)			MCWB	7.8	7.4	9.6	14.1	17.3	20.0	19.7	19.2	17.6	14.6	11.2	8.1	(20)
(21)		2%	DB	7.8	9.3	14.0	21.1	24.6	27.8	29.7	28.7	24.0	18.0	12.5	8.1	(21)
(22)			MCWB	6.1	6.6	8.3	12.5	16.0	18.5	19.2	18.5	16.4	13.4	10.1	6.6	(22)
(23)		5%	DB	6.1	7.5	11.6	18.7	22.2	25.5	27.6	26.5	21.8	16.2	11.0	6.9	(23)
(24)			MCWB	4.7	5.5	7.3	11.4	14.8	17.4	18.3	18.0	15.6	12.8	9.0	5.6	(24)
(25)		10%	DB	4.6	5.9	9.3	16.4	20.2	23.3	25.3	24.5	19.6	14.6	9.5	5.5	(25)
(26)			MCWB	3.4	4.1	6.4	10.3	13.6	16.5	17.7	17.2	14.7	12.0	7.9	4.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	8.7	10.6	14.7	18.8	21.5	21.4	20.7	18.8	15.7	11.9	8.3	(27)
(28)			MCDWB	9.7	10.4	15.5	22.2	24.8	28.6	28.1	28.0	23.5	19.0	13.5	9.5	(28)
(29)		2%	WB	6.2	7.0	9.1	13.4	17.0	19.7	20.1	19.7	17.5	14.2	10.5	6.9	(29)
(30)			MCDWB	7.3	8.5	12.9	19.8	22.7	25.4	27.1	26.2	22.1	16.9	12.1	7.9	(30)
(31)		5%	WB	4.7	5.7	7.9	12.0	15.6	18.3	19.2	18.7	16.3	13.2	9.2	5.6	(31)
(32)			MCDWB	6.0	7.2	10.8	17.5	20.8	23.9	26.0	24.9	20.5	15.6	10.7	6.8	(32)
(33)		10%	WB	3.4	4.2	6.5	10.7	14.4	17.1	18.2	17.9	15.2	12.2	8.0	4.5	(33)
(34)			MCDWB	4.4	5.8	9.0	15.7	19.1	22.3	24.1	23.2	18.8	14.4	9.4	5.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.5	5.8	7.6	10.0	10.4	10.5	10.7	10.9	9.1	6.8	4.5	4.1	(35)
(36)			MCDBR	5.2	7.1	11.9	13.8	13.4	13.9	14.7	13.8	12.3	9.0	6.2	4.6	(36)
(37)			MCWBR	4.3	5.1	7.1	7.1	6.5	6.6	6.2	5.9	6.2	5.4	4.4	3.9	(37)
(38)		5% WB	MCDWB	5.0	6.2	10.0	12.2	11.9	12.4	12.8	12.1	10.7	8.2	5.8	4.6	(38)
(39)			MCWBR	4.3	4.9	6.4	6.6	6.4	6.5	6.0	5.8	5.9	5.3	4.5	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.326	0.369	0.412	0.413	0.411	0.428	0.422	0.383	0.358	0.331	0.298	(40)	
(41)		taud	2.342	2.319	2.251	2.191	2.230	2.266	2.223	2.261	2.341	2.380	2.394	2.376	(41)	
(42)		Ebn at Noon	720	800	823	825	843	847	823	808	802	751	679	669	(42)	
(43)		Edh at Noon	65	86	111	132	132	129	133	122	101	81	62	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.79	1.53	2.53	4.08	4.94	5.28	4.99	4.41	3.15	1.80	0.84	0.57	(44)	
(45)		RadStd	0.13	0.24	0.38	0.54	0.57	0.43	0.59	0.43	0.42	0.26	0.11	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	-164	+66	+26	(46)
(47) Regional (46 neighbors)		+0.42	N/A	N/A	N/A	+0.34	+0.33	N/A	N/A	+53	+19	(47)

Nomenclature: See separate page